

Exercise - Week 7: MATH1061/7861

Duration: 30 minutes

2 Questions - 10 marks

Topics: Recursive Definitions, Solving Recurrence Relations, Set Theory
Definitions & Examples

Student Name:

Student number: |_|_|_|_|_|_|_|_|

*Your student number should be 8 digits and **not** start with an s. e.g. 41234567*

(1) (**4 marks**) Consider the sequence defined by $a_1 = 3$, $a_2 = 8$ and

$$a_n = 4(a_{n-1} - a_{n-2})$$

Use Strong Induction to prove $a_n = 2^{n-1}(n + 2)$.

(2) **(6 marks)** Let

$$A = \{n \in \mathbb{Z}_{>0} \mid n \equiv 1 \pmod{2}, n \leq 7\}, \quad B = \{n \in \mathbb{Z}_{>0} \mid n \equiv 0 \pmod{3}, n < 8\}.$$

Determine the following (no need for justification):

(a) $A \cap B$

(b) $A \cup B$

(c) $A - B$

(d) $\mathcal{P}(B)$

(e) $B \times B$

(f) Number of elements in $A \times B$.